

Total No. of Questions : 4]

SEAT No. :

PE27

[Total No. of Pages : 1

[6579] 328

T.E. (Computer) (Insem)

COMPUTER NETWORKS AND SECURITY

(2019 Pattern) (Semester - I) (310244)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Draw and explain OSI Model. [6]
b) Explain Direct Sequence Spread Spectrum. [5]
c) Explain : [4]
i) Router
ii) Switch

OR

- Q2)** a) Draw Manchester and differential Manchester code for the bit sequence : 01001011. [6]
b) Explain TCP/IP Model. [5]
c) Explain Ring and Mesh topologies. [4]

- Q3)** a) Justify answer using CRC for divisor : 1101 dividend : 100100. [6]
b) Explain IEEE 802.15 frame format. [5]
c) Give brief about design issues in DLL. [4]

OR

- Q4)** a) What are Framing, Error control and Flow control? [6]
b) Give short note on CSMA/CA, CSMA/CD. [5]
c) Explain Stop and Wait Protocol. [4]



Total No. of Questions : 4]

SEAT No. :

PC-28

[Total No. of Pages : 1

[6360]-28

T.E. (Computer Engineering) (Insem.)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester - I) (310244)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Neat diagrams must be drawn whenever necessary.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Assume Suitable data if necessary.*
- 4) *Answer Q.1 or Q.2, Q.3 or Q.4.*

- Q1)** a) Describe Bridge, Gateway and Access point. [6]
b) Explain Frequency Hopping Spread Spectrum. [5]
c) Explain LAN and MAN. [4]

OR

- Q2)** a) Draw Manchester and differential Manchester code for the bit sequence: 01001011. [6]
b) Explain TCP/IP Model. [5]
c) Explain Ring and Mesh topologies. [4]

- Q3)** a) Give brief about HDLC Protocol. [6]
b) Justify answer using CRC for divisor: 1101 dividend: 100100. [5]
c) Give brief about design issues in DLL. [4]

OR

- Q4)** a) What are Framing, Error control and Flow control? [6]
b) Give short note on CSMA/CA, CSMA/CD. [5]
c) Explain Stop and Wait Protocol. [4]

Total No. of Questions : 4]

SEAT No. :

P8558

[Total No. of Pages : 1

Oct-22/TE/Insem-528

T.E. (Computer Engineering)

COMPUTER NETWORKS AND SECURITY

(2019 Pattern)(Semester-I) (310244)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Neat Diagram must be drawn wherever necessary.*
- 2) *Figures to the right indicates full marks.*
- 3) *Assume Suitable data if necessary.*
- 4) *Answer Q1 or Q2, Q3 or Q4.*

- Q1)** a) Explain LAN, MAN, WAN. [6]
b) What are design issues of layers? Explain it. [5]
c) Explain [4]
i) Router
ii) Switch

OR

- Q2)** a) Explain Client Server and Peer to Peer Network. [6]
b) Draw Manchester and differential Manchester code for the bit sequence: 0100110 [5]
c) Explain Star and Bus topologies. [4]
- Q3)** a) Give brief about design issues in DLL. [6]
b) Differentiate between Pure and Slotted ALOHA [5]
c) Explain PPP. [4]

OR

- Q4)** a) Sender Sends code 1001101 justify receiver using Hamming code. [6]
b) Explain IEEE 802.3 frame format. [5]
c) Explain Stop and Wait Protocol. [4]



Total No. of Questions : 4]

SEAT No. :

P-5028

[Total No. of Pages : 1

[6187]428

T.E. (Insem.) (Computer Engineering)
COMPUTER NETWORKS & SECURITY
(2019 Pattern) (Semester - I) (310244)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Draw and explain OSI Model. [6]
b) Explain Direct Sequence Spread Spectrum. [5]
c) Explain i) Router ii) Switch. [4]

OR

- Q2)** a) Explain TCP/IP Model. [6]
b) Draw Manchester and differential Manchester code for the bit sequence: 01001011. [5]
c) Explain Ring and Mesh topologies. [4]

- Q3)** a) Justify answer using CRC for divisor: 1101 dividend: 100100. [6]
b) Explain IEEE 802.15 frame format. [5]
c) Give brief about design issues in DLL. [4]

OR

- Q4)** a) What is sliding window protocol? How it works? [6]
b) Explain CSMA in detail. [5]
c) Explain PPP. [4]

